

Department of Public Institutions and Digital Governance

Digital Governance Branch Existing Applications and Infrastructure

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1. Overview

The Digital Government Branch (DGB) in UNDESA is responsible for evaluating how Member States utilize ICT resources to deliver public services at both the national and local levels. In addition, the branch supports capacity-building efforts to advance e-government initiatives, promoting inclusive, efficient, and transparent public administration through digital transformation.

These efforts culminate in the production of the globally recognized E-Government Development Index (EGDI), which has played a key role in enabling Member States to conduct comparative analyses of their e-governance performance and gain valuable insights into emerging trends in the digital governance landscape.

To achieve these objectives and effectively manage and disseminate the resulting data, the branch relies on enterprise applications that support its operations and ensure the timely delivery of crucial information.

1.1 Objectives of the Applications Used: egovsurvey and egovkb

To support its mandate of evaluating e-government development across all UN Member States, the Digital Government Branch (DGB) in UNDESA relies on two key applications: egovsurvey and egovkb. These platforms play a critical role in facilitating data collection, analysis, and dissemination related to global e-government performance.

EGOVSURVEY (Online Assessment Tool) Objectives:

- Serve as the primary platform for collecting structured data from national and municipal portals through volunteer-administered questionnaires, while facilitating the systematic assessment of these portals to ensure consistency in evaluating e-government data.
- Provide a centralized environment for data validation, expert review, ensuring that the data used for the E-Government Development Index (EGDI) is accurate, comparable, and timely.
- Enable tracking of trends over time, supporting longitudinal analysis of digital government progress across regions and income groups.

EGOVKB (E-Government Knowledgebase) Objectives:

- Act as a public-facing knowledge repository, providing open access to the complete historical datasets, reports, and analysis derived from the United Nations E-Government Surveys.
- Facilitate comparative analysis by allowing users to explore and compare countries' e-government performance across various indicators.
- Support knowledge exchange by making best practices, trends, and regional insights accessible to policymakers, researchers, and practitioners worldwide.
- Ensure timely dissemination of survey results, updates, and thematic reports to stakeholders in a user-friendly format.
- Serve as a capacity-building tool, helping governments identify strengths, gaps, and opportunities for enhancing their digital government strategies.

2. DGB Data Overview

The E-Government Development Index (EGDI) and the Local Online Service Index (LOSI) follow a structured, multi-dimensional data model designed to capture various aspects of e-government development at both the national and local levels.

For EGDI, the data structure is built around three core components, each sourced from different data streams:

- Online Services Index (OSI) – Data collected directly by UN DESA, through volunteer assessments of national government portals and key ministry websites. Volunteers evaluate a structured set of criteria covering service availability, content provision, technology features, and e-participation mechanisms.
- Telecommunications Infrastructure Index (TII) – Data sourced from the International Telecommunication Union (ITU), covering key infrastructure indicators such as internet usage, mobile and broadband subscriptions, and service affordability.
- Human Capital Index (HCI) – Data sourced primarily from UNESCO and UNDP, covering literacy rates, gross enrollment ratios, expected years of schooling, and mean years of schooling. The 2024 Survey also introduced E-Government Literacy (EGL), which is derived from features assessed directly through government portals.

Each component is normalized and combined into the composite EGDI score, providing a comparative measure of e-government development across all UN Member States.

For LOSI, the structure is similar but applied at the city level, with data collected directly through volunteer assessments. LOSI focuses on:

- Institutional Framework (IF)
- Services Provision (SP)
- Content Provision (CP)
- Technology (TEC)
- E-Participation (EPI)
- E-government literacy (EGL)

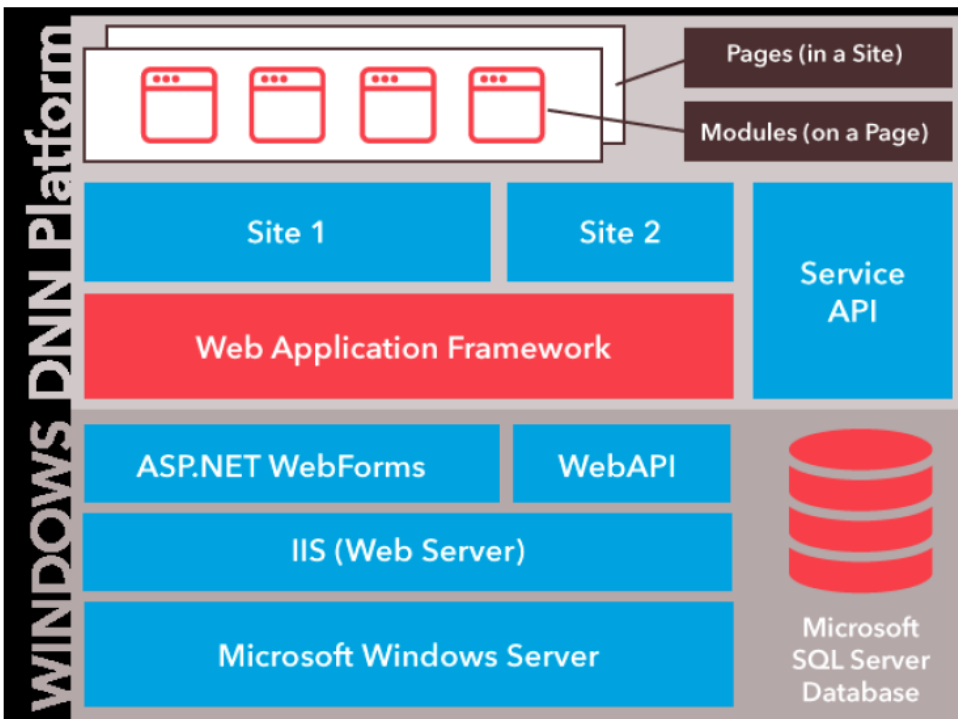
Both indices rely on a combination of binary scoring (yes/no) and weighted scoring for more complex elements, such as transactional service maturity and participatory mechanisms. Final EGDI and LOSI values are normalized on a scale of 0 to 1, enabling both global benchmarking and regional comparisons.

3. System Architecture (Shared)

3.1 Application Architecture

Both egovsurvey and egovkb use DNN open-source platform technology. Both applications are on DNN version 9.6.1.

Figure 1: Application Architecture



3.2 Servers Architecture

The databases for egovsurvey and egovkb are located at the NYVM2501 server and the application files are hosted on NYVM2500 server, as described at the diagram below:

Figure 2: Server Architecture



3.3 Access to the Hosting Environment

All servers can be accessed through remote desktop protocol on windows. Both the web server and database server are still on Microsoft Active directory (UNHQ) and **NOT** on Microsoft Azure AD. Therefore, to gain access, a localhost or domain admin must configure a domain account on both servers. External Remote access to administrative modules on the applications requires UNHQ VPN.

Figure 3: Remote Access to the web servers

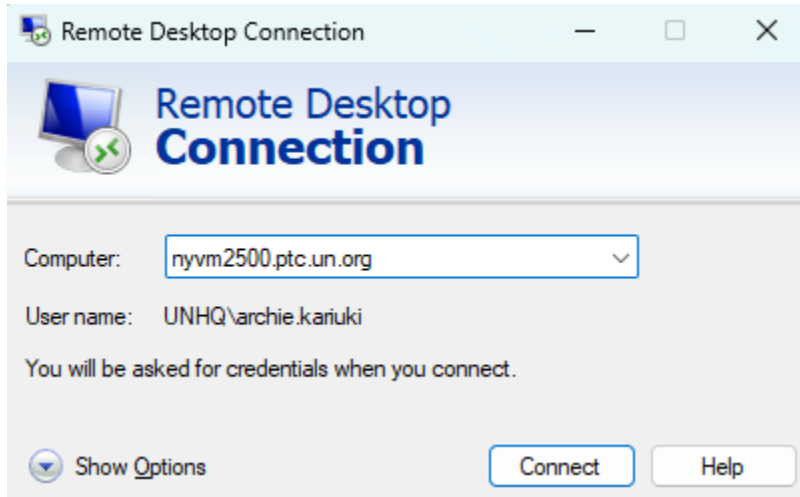


Figure 4: Remote Access to the database servers

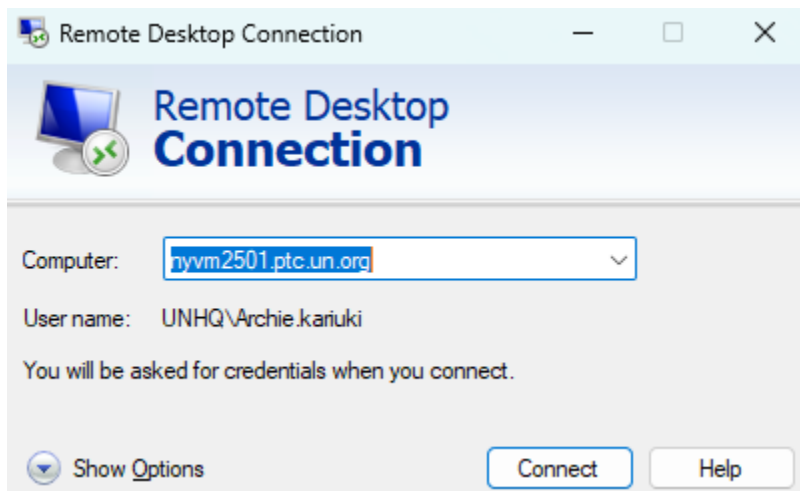
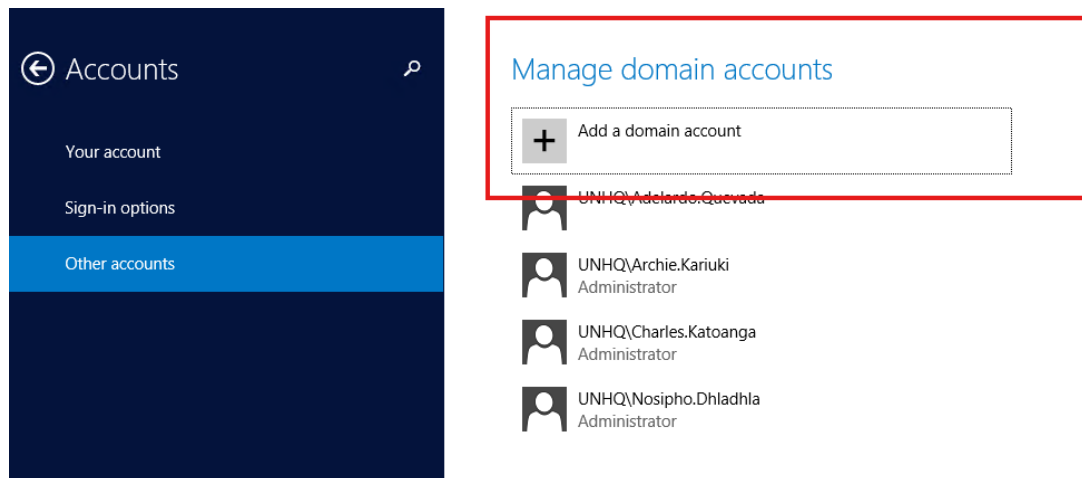


Figure 5: Creating Domain accounts



4. EGOVSURVEY

EGOVSURVEY is a web application designed to support the evaluation process of national and local e-government portals. It serves as a core tool in the preparation of the UN E-Government Survey, providing a structured environment for volunteers and researchers to assess government portals across all 193 UN Member States as well as selected cities.

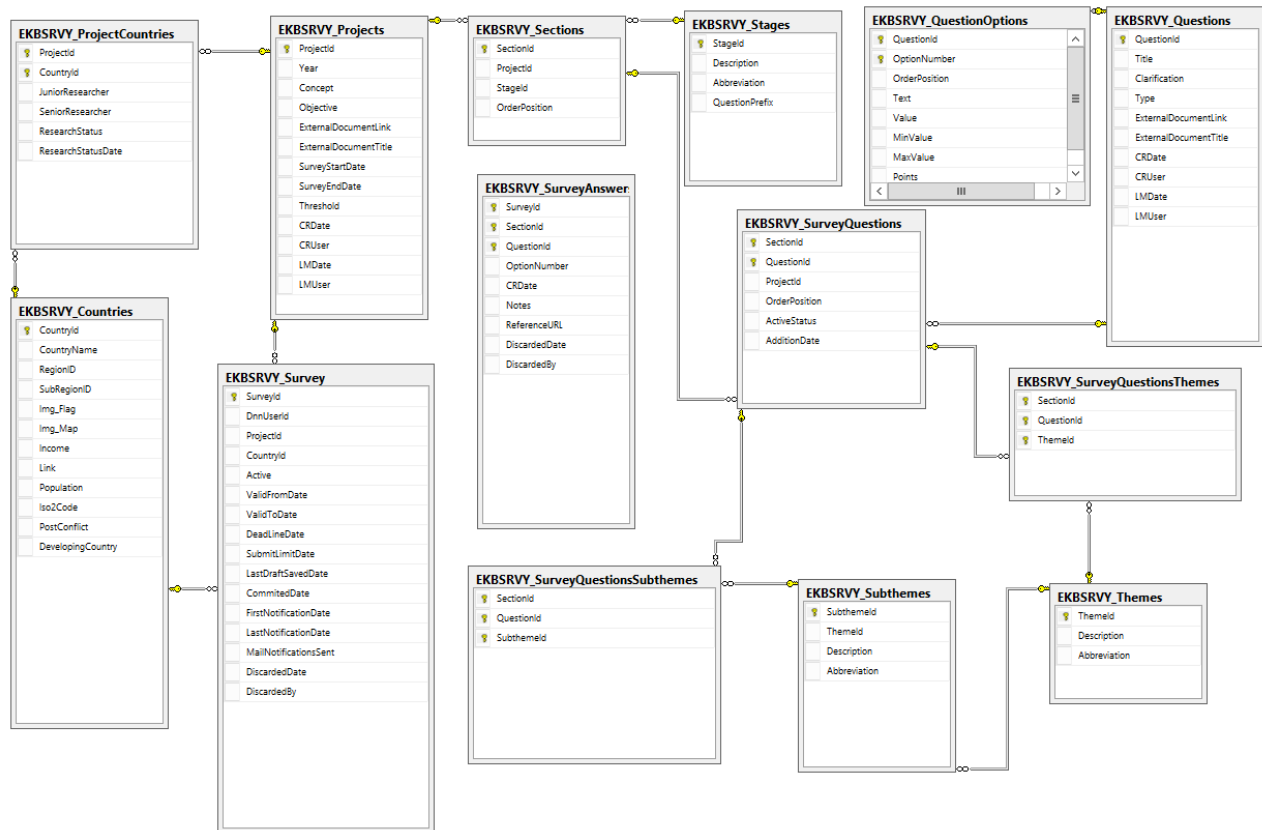
The platform hosts predefined evaluation criteria aligned with the E-Government Development Index (EGDI) and Local Online Service Index (LOSI) methodologies, allowing users to record observations, assign scores, and document supporting evidence such as screenshots and URLs directly within the system.

EGOVSURVEY also facilitates multi-user access, enabling volunteers, supervisors, and data managers to collaborate throughout the assessment process, while providing an audit trail to track changes and ensure data transparency.

4.1 System Components

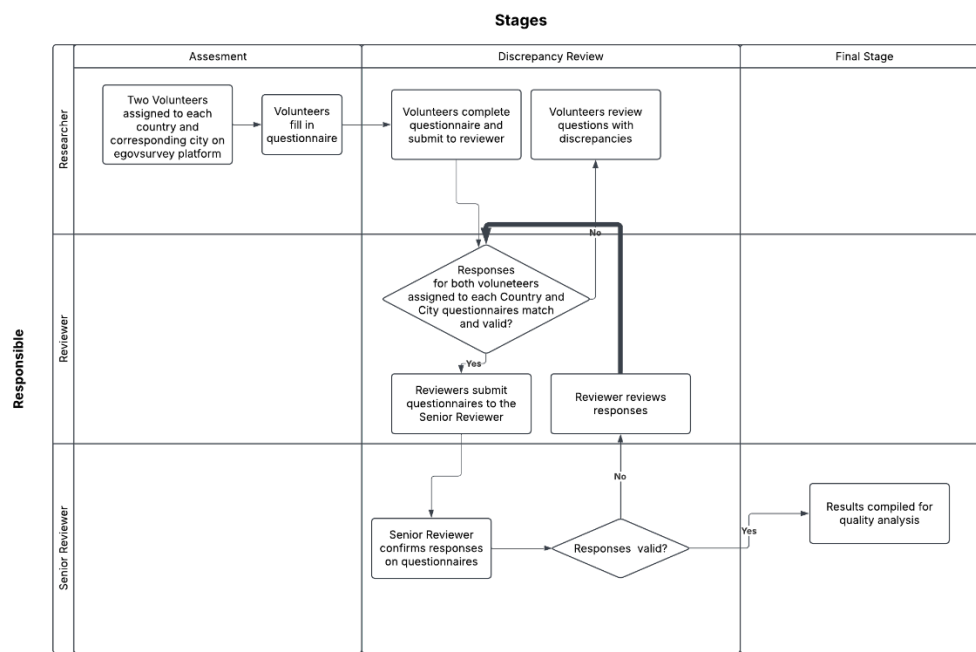
4.1.1 Database Design

Figure 6: Egovsurvey Database Schema



4.2 Process Workflows

Figure 7: Assessment Workflow



4.2.1 Platform access

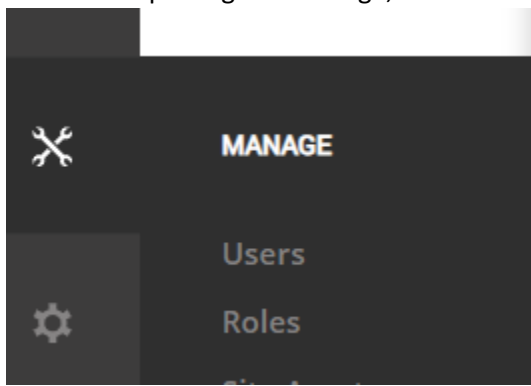
EGOV SURVEY has two login screens i) one for regular users such as UNVs, Junior and Senior Reviewers and Managers and ii) a restricted admin login screen that can be only accessed within UN premises. Admin Login is only needed for adding a new user to the platform (please see separate manual for adding new users), any other project management can be done with “Manager” role through “USER LOGIN” below.

- **USER LOGIN:** for UNVs, Junior and Senior Reviewers and Managers:
 - <https://publicadministration.un.org/egovsurvey/>
- **ADMIN LOGIN:** for Administrators (only accessible from UN on-site network)
 - <https://publicadministration.un.org/egovsurvey/LoginPage/LoginAdmin.aspx>

EGOV SURVEY is also hosted on **NYVM2500** web server, and its database (“SurveyEgov”) resides on **NYVM2501** server.

4.2.2 Configuring user accounts for volunteers

- Access the admin back-end through the link <https://publicadministration.un.org/egovsurvey/LoginPage/LoginAdmin.aspx>
- On the side panel go to Manage, and then Users as seen below :



Add the new user details and save. The new user will be created with the role **Registered User** which will grant access to the survey questionnaires.

Users Remove Deleted Users Add User

Show: Authorized Search Users

NAME	EMAIL	JOINED
First Name:	Last Name:	
User Name:	Email Address: TempHost	
Authorized: ☒ On	Random Password: ☐ Off	
Password: ***** fair	Confirm Password:	
☐ Send An Email To New User		
Cancel Save		

To add roles to any user, go to Manage, and then Users. Select the desired user and click on the shield icon to add roles to the particular user. After clicking the icon, type in the role you want to assign on the search box under the Roles section.

Users Remove Deleted Users Add User

Show: Authorized Search Users

NAME	EMAIL	JOINED										
Monina Tran tran.monina	monina.tran@un.org	01/10/2025										
ROLES ☒ Send Email Begin typing to add a role to this user. + Add												
<table border="1"> <thead> <tr> <th>ROLE</th> <th>START</th> <th>EXPIRES</th> </tr> </thead> <tbody> <tr> <td>Registered Users</td> <td>-</td> <td>-</td> </tr> <tr> <td>Subscribers</td> <td>-</td> <td>-</td> </tr> </tbody> </table>				ROLE	START	EXPIRES	Registered Users	-	-	Subscribers	-	-
ROLE	START	EXPIRES										
Registered Users	-	-										
Subscribers	-	-										
Test User testuser	archie.kariuki@un.org	09/11/2024										
Archie Kariuki kariuki	archie.kariuki@un.org	05/15/2023										

4.2.3 General Configuration

- Can be accessed from “Management” tab:
<https://publicadministration.un.org/egovsurvey/Management/EGovernmentSurvey.aspx>
- General configuration works like adding a new project, managing question set and country/city set can be done here on the upper part of “Management” tab.



- Currently cities are added from “*Countries&Regions*” menu and they must have a naming standard that starts with the prefix of its country’s code followed by a dash character ‘-’ and space in between. For example, Dubai was added as “AE - Dubai”.

Adding a new project

- Click “Projects”
- On “Project Manager” page, duplicate the existing projects to create new projects. **Don’t use “Add New Project” button.**
- Use “2022 UN E-government Survey” project for OSI related new projects and use “2022 LOSI E-Government Survey” for LOSI related projects (including LOSI pilots).

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UN Public Administration Programme
Division for Public Institutions and Digital Government
UN Department of Economic and Social Affairs (UNDESA)

Home | Management | Surveys | Reviewers | yusuf.eren | Logout

PROJECT MANAGER

Show 10 entries Back to control panel

Year	Title	Date limits	% Threshold	Created	Actions
2021	LOSI_Pilot_2021 - MOCK	From: 7 Jun 2021 To: 30 Jul 2021	20.000000	On: 14 Jun 2021 By: TempHost	  
2022	PILOT_2022 UN E-government Survey	From: 6 Jul 2021 To: 16 Jul 2021	20.000000	On: 6 Jul 2021 By: TempHost	  
2022	PILOT_2022 LOSI E-Government Survey	From: 12 Jul 2021 To: 30 Jul 2021	20.000000	On: 12 Jul 2021 By: yusuf.eren	  
2022	2022 UN E-government Survey	From: 14 Jul 2021 To: 31 Oct 2021	20.000000	On: 14 Jul 2021 By: TempHost	  
2022	2022 LOSI E-Government Survey	From: 14 Jul 2021 To: 31 Oct 2021	20.000000	On: 14 Jul 2021 By: TempHost	  
2022	2022 Test UN E-Government Survey	From: 1 Dec 2020 To: 31 Dec 2022	20.000000	On: 30 Dec 2020 By: TempHost	  
2022	PILOT 2022 UN E-government Survey	From: 26 Jan 2021 To: 30 Sep 2021	20.000000	On: 26 Jan 2021 By: TempHost	  

Showing 11 to 17 of 17 entries

[Add new project](#)

First Previous 1 2 Next Last

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- Choose *Year* and type a unique *Title*. Use current date as *Start Date* and make sure to have like 9 months for the duration of project so choose *End Date* as 9 months after the *Start Date*.
- Keep *% Threshold* as 20.
- Make sure Project Indexes, Project stages with Survey Questions, Survey Countries and Survey URLs and Surveyors are selected(keep *Copy only active surveyors* unselected). *Survey Valid From* should be the day UNVs will start working on Surveys. *Survey Valid to, Submitting deadline* and *Submit limit date* should be the deadline for surveyors to complete their assessments (These dates can be changed later). Please see the screenshot below.

2021	LOSI_Pilot_2021_Palestine	From: 31 Oct 2021 To: 31 Oct 2021	20.000000	On: 12 Aug 2021 By: yusuf eren	  
2021	LOSI_Pilot_2021_UK	From: 11 Jan 2021 To: 12 Apr 2021	20.000000	On: 11 Jan 2021 By: TempHost	  
2021	LOSI_Pilot_2021_Brazil_Jordan	From: 7 Jun 2021 To: 30 Jul 2021	20.000000	On: 7 Jun 2021 By: TempHost	  

Showing 1 to 10 of 17 entries

First Previous 1 2 Next Last

Duplicate a project

Year of the surveys: 2022

Title: 2022 UN E-government Survey duplicate test

Objective: 2022 UN E-government Survey

External document link:

External document title:

Start date of the surveys: 7/18/2022

End date of the survey: 12/31/2022

% Threshold: 20.000000

Select the following options to determine what is to be copied to the new project

- ☒ Project indexes
- ☒ Project stages (sections)
 - ☒ Survey questions
- ☒ Survey countries
 - ☒ Survey URLs
- ☒ Surveyors

Necessary information for surveyors

☐ Copy **only** active surveyors

Survey valid from: 7/18/2022

Survey valid to: 12/31/2022

Submitting deadline: 12/31/2022

Submit limit date: 12/31/2022

Save Cancel

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- Click Save

4.2.4 Project Configuration

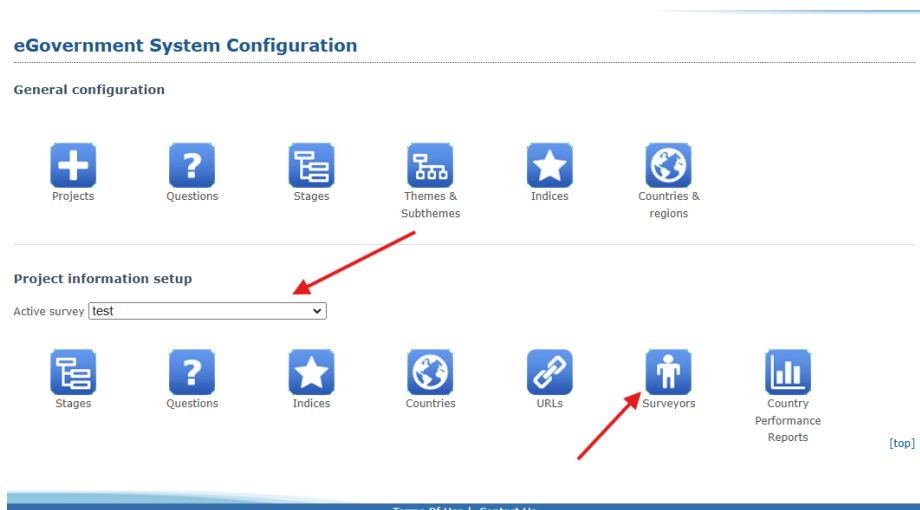
- Project specific operations such as adding/removing questions, countries/cities and surveyors to a certain project can be done from the bottom half of the “Management” page.



- Use **Surveyors** to assign new assessments to the surveyors who should be added as new user beforehand to appear on the list. (Refer to the “ADMIN LOGIN” in the beginning of this document and see separate manual for adding new users.)

4.2.3 Assigning users to surveys

Under the management tab, from the dropdown box, select your preferred Survey. Then click on the Surveyors icon to view the page for assigning users to surveys.



On the Surveyors page, click on the **Add a new Surveyor**

On the open Questions page, click on the Add a question button.

»Features such as Facebook, including the 'Like' button, and Twitter. Look for different social networks according to the country assessed i.e. Sina Weibo (China), Odnoklassniki (Russia). See http://www.techinasia.com/2013-china-top-10-social-sites-infographic/ for further information.	Choice	2 Jul 2019 Jaemin Kim	2 Jul 2021 yusuf eren	✗
#008 Is there a "contact us" feature, e.g. email, phone number (one-way), or address? »The "contact us" feature can be found under a variety of labels. However, as long as one finds a method of contact, this feature should be scored "Yes".	Choice	2 Jul 2019 Jaemin Kim	2 Jul 2021 yusuf eren	✗
#009 Is there a "live chat" support functionality with a person? »This covers real-time interaction through "click to chat", wechat, WeChat, Whatsapp, text messaging, and other live support functions.	Choice	2 Jul 2019 Jaemin Kim	29 Jun 2023 Zhenye Pan	✗
#010 Does the portal(s) offer AI-enabled chat-bot functionality? »Score: 0-for NO, 1-if AI enabled chatbot functionality is available, 2: If it supports Generative AI.	Multiple choice	29 Jun 2023 Zhenye Pan	23 Aug 2023 Zhenye Pan	✗

Showing 1 to 10 of 1,280 entries

[Add new question](#)

First Previous 1 2 3 4 5 Next Last

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On the same page below, fill in the details of the question and select the question type.

Adding a new question...

Question

Guideline

Type

External document link

External document title

To assign the question or change the order of how questions appear on the questionnaire, you first must assign stages to categorize the questions. To assign stages, in the Project Information set-up section, select the survey to configure in the drop-down menu, and then click on the Stages Icon.

eGovernment System Configuration

General configuration

+

Projects

?

Questions

☰

Stages

🔗

Themes & Subthemes

★

Indices

🌐

Countries & regions

Project information setup

Active survey

☰

Stages

?

Questions

★

Indices

🌐

Countries

🔗

URLs

👤

Surveyors

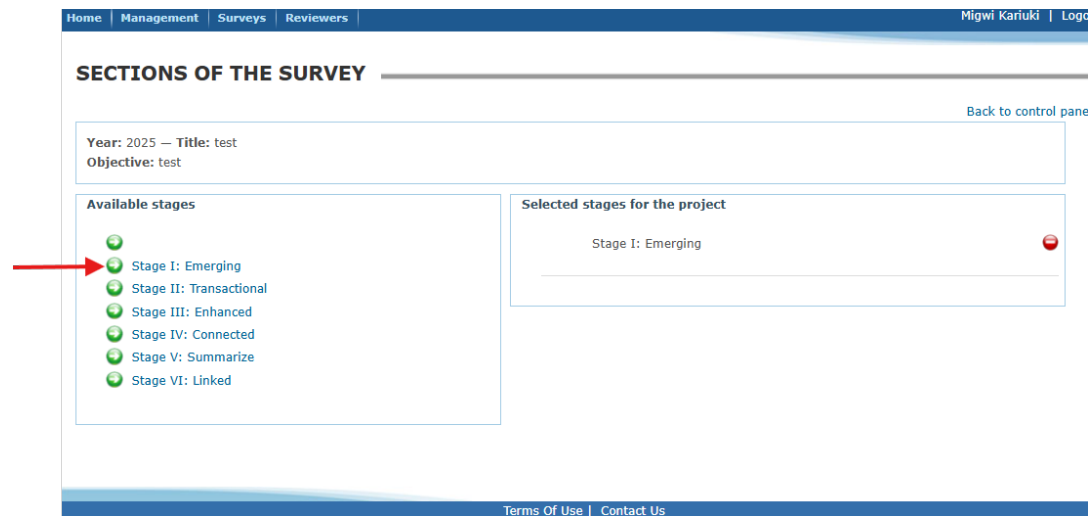
📊

Country Performance Reports

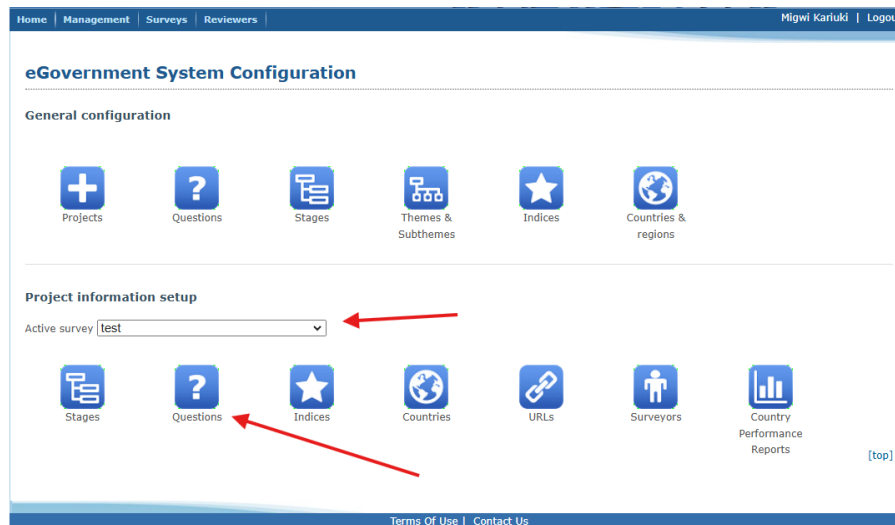
[top]

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Click on the green arrow icon adjacent to the stage of choice to add the corresponding stage.



After adding the stage, go back to Management tab on the top navigation menu, and in the Project Information setup section, make sure the survey of choice is selected in the dropdown menu, and then click on the Questions icon.



On the open Survey questions page, in the the Available Questions section, select the questions that you want to include in the survey by clicking on the checkbox adjacent to the question and make sure the stage is selected in the same panel.

Available questions

[1280 questions]

Show 10 entries

☒ #001 Is there at least one national government portal?
 ☐ #002 Can any one of the government portal(s) be found on the first results page of any search engine typically used in that country?
 ☐ #003 Is there a search feature in its homepage?
 ☐ #004 Is there an advanced search option?
 ☐ #005 Is there a sitemap (or index)?
 ☐ #006 Is there a 'Help' feature or 'Frequently Asked Questions (FAQs)' section?
 ☐ #007 Are there any social networking feature(s)?
 ☐ #008 Is there a "contact us" feature, e.g. email, phone number (one-way), or address?
 ☐ #009 Is there a "live chat" support functionality with a person?
 ☐ #010 Does the portal(s) offer AI-enabled chat-bot functionality?

Showing 1 to 10 of 1,280 entries

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[2](#)
[3](#)
[4](#)
[5](#)
[Next](#)
[Last](#)

Select a section

Stage I: Emerging

Select themes & subthemes

☐ Digital Divide; Vulnerable groups (DDVG)
 ☐ Low income (I)
 ☐ Environmental (ENV)
 ☐ e-Participation (EP)
 ☐ e-Consultation (eC)
 ☐ E-Consultation (EP-C)
 ☐ E-Decision-Making (EP-D)
 ☐ E-Information (EP-I)
 ☐ Expanding Usage (EU)
 ☐ Multi-Channel Service Delivery (MCSD)
 ☐ Open Government / Data (OGD)
 ☐ Test.Stage.1 (SG1)
 ☐ Whole of Government (WoG)

Assigned questions

[0 questions]

Export data to XML file

Then click on Assign the checked question(s) to add the question onto the Survey

functionality?

Showing 1 to 10 of 1,280 entries

[First](#)
[Previous](#)
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[3](#)
[4](#)
[5](#)
[Next](#)
[Last](#)

Select a section

Stage I: Emerging

Select themes & subthemes

☐ Digital Divide; Vulnerable groups (DDVG)
 ☐ Low income (I)
 ☐ Environmental (ENV)
 ☐ e-Participation (EP)
 ☐ e-Consultation (eC)
 ☐ E-Consultation (EP-C)
 ☐ E-Decision-Making (EP-D)
 ☐ E-Information (EP-I)
 ☐ Expanding Usage (EU)
 ☐ Multi-Channel Service Delivery (MCSD)
 ☐ Open Government / Data (OGD)
 ☐ Test.Stage.1 (SG1)
 ☐ Whole of Government (WoG)

Assign the checked questions

The added questions will now appear on the right panel under **Assigned questions**.

Year: 2025 — Title: test

Objective: test

Available questions

[1278 questions]

Show 10 entries

☐ #003 Is there a search feature in its homepage?
 ☐ #004 Is there an advanced search option?
 ☐ #005 Is there a sitemap (or index)?
 ☐ #006 Is there a 'Help' feature or 'Frequently Asked Questions (FAQs)' section?
 ☐ #007 Are there any social networking feature(s)?
 ☐ #008 Is there a "contact us" feature, e.g. email, phone number (one-way), or address?
 ☐ #009 Is there a "live chat" support functionality with a person?
 ☐ #010 Does the portal(s) offer AI-enabled chat-bot functionality?

Assigned questions

[2 questions]

Stage I: Emerging

I- 1

#001 Is there at least one national government portal?

Themes & Subthemes:

I- 2

#002 Can any one of the government portal(s) be found on the first results page of any search engine typically used in that country?

Themes & Subthemes:

Export data to XML file

Assigned questions [2 questions]

Stage I: Emerging

I- 1 #001 Is there at least one national government portal?
 **Themes & Subthemes:**  

I- 2 #002 Can any one of the government portal(s) be found on the first results page of any search engine typically used in that country?
 **Themes & Subthemes:**  

[Export data to XML file](#)

5.1 System Components

Figure 8: EGOVKB Database schema

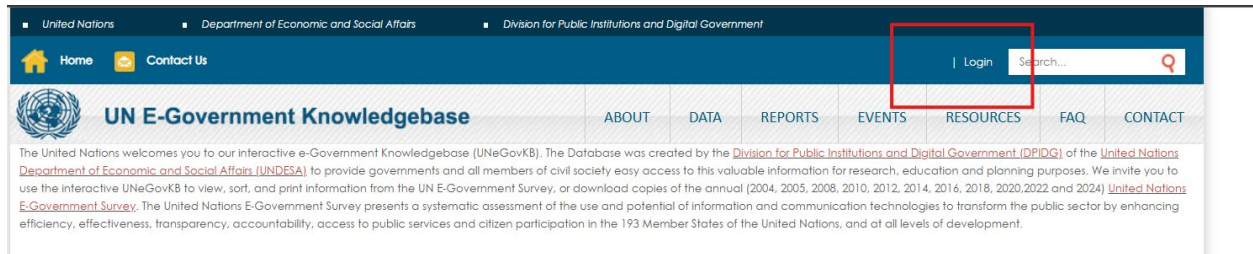
[illegible]

5.2 Process Workflows

5.2.1 Platform access

The platform is public facing and can be accessed using the link <https://publicadministration.un.org/egovkb>

To edit content on the application click on the login button above the top navigation menu as seen below.



The screenshot shows the top navigation bar of the UN E-Government Knowledgebase. The 'Login' button is highlighted with a red box. Below the navigation bar, the main header includes the UN logo and the text 'UN E-Government Knowledgebase'. A brief description of the platform is provided, followed by a 'User Log In' section.

User Log In

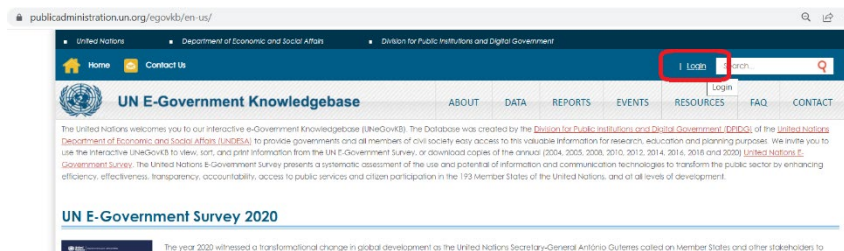
Username:

Password:

☐ Remember Login

5.2.2 Updating EGOVKB data with new Survey data using “EGOV DATA ADMIN” module

1. Login to EGOVKB (<https://publicadministration.un.org/egovkb>) with admin credentials.



2. Find “EGOV DATA ADMIN” module under DNN control panel’s *Manage* menu at the left hand side of screen and click to it.

UN EGOVKB | United Nations > Home x +

← → ↻ publicadministration.un.org/egovkb/en-us/

United Nations Department of Economic and Social Affairs

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UN E-Government Knowledge

The United Nations welcomes you to our interactive e-Government Knowledge Department of Economic and Social Affairs (UNDESA) to provide government view, sort, and print information from the UN Nations E-Government Survey presents a system, efficiency, accountability, access to public service

Survey 2020

Year 2020 witnessed a transformational change start a decade of delivery and action for people systems of digital government around the world identifying their strengths, challenges and opportunities services to all and to bridge the digital divide. The Survey has become an indispensable temporary research on e-government. The role of e-government, both in its conventional divides to the fore, especially among the poorest

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MANAGE

- Users
- Roles
- Site Assets
- Global Assets
- Themes
- Sites
- Site Groups
- Admin Logs
- EGOV DATA ADMIN**
- Open Url Rewriter
- Security Analyzer

- Click "E-Index Data" under "Manage Data" to start adding/updating data related to all Index Types i.e. EGDI/OSI/HCI/TII and EPI.

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Division for Public Institutions

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UN E-Government Knowledgebase

ABOUT

DATA

EGOV - Admin

Visible by Administrators only.

Manage Data

E-Index Data

Countries

Regions

Cities

Region Analyses

Index Types

4. Choose “Year” and “Index Type” from the list (will be done for each Index Type separately) and click “Edit Data” button.

EGOV - Admin

Visible by Administrators only.

Admin

E-Index Data

Choose a Year and Index Type:

The screenshot shows a web form for 'E-Index Data'. It includes a 'Year' dropdown menu set to '2022', an 'Index Type' dropdown menu set to 'E-Government Development Index', and an 'Edit Data' button. Red boxes and numbers highlight these elements: a red box around the 'Year' dropdown is labeled '1'; a red box around the 'Index Type' dropdown is labeled '2'; and a red box around the 'Edit Data' button is labeled '3'.

Year 2022 ▼ 1

Index Type E-Government Development Index ▼ 2

Edit Data 3

5. Fill the Value/Rank pairs for each country and click "Save". This step can be done at once or part by part as long as it is saved each time. Click "Admin" to go back to main menu of "Manage Data" and "E-Index Data Points" for previous step.

EGOV - Admin

Visible by Administrators only.

Admin

E-Index Data Points

2022

E-Government Development Index

Save

Country	Value	Rank
Afghanistan	0.27100	184
Albania	0.74130	63
Algeria	0.56110	112
Andorra	0.71770	76
Angola	0.37890	157
Antigua and Barbuda	0.61130	99
Argentina	0.81980	41
Armenia	0.73640	64
Australia	0.94050	7

NOTE: All the indices, namely EGDI, OSI, HCI, TII, EPI and LOSI must be populated in order for each individual Country/City pages to work properly. For example Chile's country information page below will only be showing full data after all the indices listed above are updated. Same applies to city pages as well, see example city page for Madrid.

<https://publicadministration.un.org/egovkb/en-us/Data/Country-Information/id/35-Chile>

<https://publicadministration.un.org/egovkb/en-us/Data/City/id/68-Madrid/dataYear/2022>